

1. Introduction

1.1. Clinical and Histopathological Foundations

Histology is the microscopic study of the structure and composition of biological tissues and has been foundational to medical diagnosis since the nineteenth century [1] [2]. By preparing thin sections of tissue and staining them with dyes that selectively color different cellular components, histologists render the invisible architecture of organs and tissues visible under a microscope [2].

Histopathology extends this discipline to the study of diseased tissues, serving as the gold standard for diagnosing a wide range of conditions, from infections to malignancies [3] [4]. **The word histopathology was first introduced in a book entitled "On the Nature and Structure Characteristic of Cancer" by Johannes Muller in 1838 [4].** When abnormal changes in cell morphology, tissue architecture, or staining patterns are identified, they provide crucial evidence for determining the nature, extent, and severity of disease.

Among the most consequential applications of histopathology is cancer diagnosis and staging. Cancer is the uncontrolled proliferation and spread of abnormal cells; it remains a leading cause of death worldwide [5]. A critical step in determining a patient's prognosis and treatment plan is assessing whether the tumor has metastasized, that is, disseminated beyond its primary site to distant organs and tissues, most commonly through the lymphatic system [6]. Lymph nodes, which function as filters of tissue fluids and as sites of lymphocyte production, are therefore routinely examined histopathologically to detect metastatic deposits [7]. In breast cancer, the histopathological assessment of sentinel lymph node biopsies directly underpins the TNM staging system and informs treatment decisions [8].

To perform these diagnostic assessments, standard histopathological examination relies fundamentally on microscopy and tissue staining. The light microscope, first assembled by Zacharias Jansen in the sixteenth century and perfected by Leeuwenhoek in 1673, remains the primary tool for routine diagnostic work, with resolution governed by the ratio between the wavelength of light and the width of the diaphragm opening [4]. More advanced instruments, including scanning electron microscopes (SEM), transmission electron microscopes (TEM), cryo-electron microscopes (Cryo-EM), fluorescence microscopes, and phase-contrast microscopes, extend resolution to the ultrastructural and even atomic level for specialized applications [4].

The most widely used stain in histopathology is hematoxylin and eosin (H&E), first described in 1875–1878, in which hematoxylin colors cell nuclei blue and eosin stains the cytoplasm and extracellular matrix pink, enabling pathologists to evaluate tissue architecture and cellular morphology [1] [14]. Immunohistochemistry (IHC) further advances diagnostic capability by using labelled antibodies to detect specific antigens within tissue sections, enabling the identification of particular cell types, tumor markers, and prognostic biomarkers such as HER2 amplification in breast cancer [15], [16]. However, the preparation of high-quality slides is itself a complex, multi-step process, involving fixation, processing, embedding, sectioning, staining, and mounting, each step of which can introduce artefacts that alter tissue morphology and potentially compromise diagnostic accuracy [3].

Beyond the physical preparation of the tissue, the subsequent manual examination of whole-slide images is time-consuming, subjective, and prone to inter-observer variability for instance, breast cancer subtyping by expert pathologists has shown accuracy rates as low as 57.08%, and prostate cancer grading inter-observer agreement has been reported at only 57.90% [9].

Given that cancer diagnosis depends on such subtle morphologic distinctions, histopathologic interpretation is usually performed by specialist pathologists, and second-opinion review is often warranted when the diagnosis may alter staging or treatment. In a large breast-biopsy study, overall agreement between individual pathologists and a consensus reference was 75.3%, with the weakest agreement in atypia and ductal carcinoma in situ [10]. Specialist review also amended 16.5% of pediatric cancer diagnoses [11], and in early rectal cancer discordant histopathologic assessment changed treatment in 11% of patients [12].

A separate review of anatomic pathology error detection reported that interpretive error rates can range from 1.2 to 50 errors per 1,000 cases, and it highlighted double-reading, case conferences, and consultation as the traditional quality-control triad [13].

1.2 Digital Pathology and Domain Challenges

While traditional quality-control relies on human consultation and manual glass-slide review, the integration of digital pathology has introduced a new technological layer to this workflow, allowing whole-slide images (WSIs) to be viewed on a computer screen for remote consultation, annotation, and computational analysis. Open-source tools used in this workflow include QuPath,

HistoQC, and HistomicsTK [17]. However, moving glass slides to the computational domain introduces severe infrastructure bottlenecks [18]. The sheer size of Whole-Slide Images (WSIs), with the CAMELYON dataset alone comprising 1,399 WSIs and consuming roughly 2.95 terabytes of storage, prevents them from being loaded into standard GPU memory in their entirety [19], [20], [21]. This constraint has necessitated patch-based processing paradigms, in which WSIs are tiled into small patches (typically 96×96 or 256×256 pixels), classified individually, and then aggregated into slide-level or patient-level predictions [22], [23].

To ensure these patches are of sufficient quality, preprocessing steps, including tissue detection, background removal, and stain normalization, are essential [24]. A primary challenge in this stage is "domain shift," where model performance degrades on images from unseen institutions due to variations in staining protocols, reagent brands, tissue preparation, and scanner hardware. Resolving this institutional color and textural variability is critical, yet the optimal interaction between explicit normalization pipelines and modern neural networks remains an active area of investigation. Several classical stain normalization methods, such as those proposed by Macenko, decompose H&E images into stain vectors to map them to a reference template [25], [26]. (move to lit review)

1.3 Evolution of Computational Algorithms

Early computational approaches to histopathology relied on classical machine learning and manually engineered features, such as Local Binary Patterns (LBP) and Scale-Invariant Feature Transform (SIFT), classified using Support Vector Machines (SVMs) or Random Forests [28].

While these methods demonstrated the feasibility of automated detection in breast and colorectal cancers, they were heavily dependent on human intuition for feature selection and often failed to capture the full heterogeneity of tumor microenvironments [29]. When compared directly on breast cancer datasets, deep learning approaches achieved superior accuracies of 94–98% for binary classification, compared to 85–89% for classical methods [30]. This formal shift toward Convolutional Neural Networks (CNNs) allowed models to learn hierarchical feature representations directly from raw pixel data, eliminating the need for handcrafted engineering [31]. A landmark study demonstrated through the CAMELYON16 challenge that the top-performing deep learning algorithm achieved an AUC of 0.994 for slide-level metastasis classification, significantly outperforming a panel of eleven pathologists (mean AUC 0.810). (move to lit review)

Progress was further accelerated by transfer learning, where fine-tuned architectures like VGG16 and ResNet50 significantly outperformed models trained from scratch [32] [33].

by transfer learning; fine-tuned architectures like VGG16 yielded 92.60% accuracy, significantly outperforming models trained from scratch [32], while pre-training on domain-specific histopathology data showed even greater gains [33].(move to lit review)

More recently, self-supervised learning and pathology-specific foundation models, including CTransPath, UNI, CONCH, GigaPath, and the Virchow family, pretrained on millions of whole-slide images, have pushed the state of the art further, achieving strong performance across diverse tasks with minimal labelled data. Independent benchmarking across 13 cohorts found CONCH and Virchow2 to be the top-performing models. At the slide level, Multiple Instance Learning (MIL) frameworks aggregate patch-level features into whole-slide predictions, enabling clinically relevant classification without exhaustive pixel-level annotation. A recent systematic review covering 73 studies found that 86% of computational lymph node analysis work focused on metastasis identification, with multi-scale models mimicking multiple microscopy zooms showing the most promise [34], [35].(move to lit review)

Despite these advances, current pipelines still face substantial bottlenecks regarding institutional stain variation, model miscalibration, and a distinct lack of rigorous external validation across disparate clinical centers. Recent reviews note that task-specific AI models depend on large annotated training sets, which is a rate-limiting factor for pathology AI development, and that even foundation-model pipelines still face clinical integration and trust barriers [36].(move to lit review)

1.4 Problem Statement

Histopathological diagnosis remains the standard for cancer classification, yet it is still vulnerable to interobserver variation and specialist-dependent interpretation. Digital pathology and AI may improve reproducibility, but their use is limited by stain variation, generalizability, and interpretability concerns. This project therefore examines whether a preprocessing-aware foundation-model pipeline can improve the robustness of histopathology-based cancer diagnosis across centers.

1.5 Aims and Objectives

1.5.1 Aims

This project aims to build and evaluate a reproducible, cross-domain histopathology classification framework for lymph-node metastasis detection on H&E patch images. Using PatchCamelyon (PCam) and CAMELYON17 as complementary benchmarks, it tests whether a modern pathology foundation model (Virchow2) is robust enough to transfer across centers without careful preprocessing, or whether quality-aware, stain-standardized preprocessing remains necessary for reliable performance when scanners, staining, and lab workflows differ. The work is organized around bidirectional external validation (train on one dataset, test on the other and vice versa), controlled comparison of stain-handling strategies, and evaluation beyond accuracy; including calibration, uncertainty, and structured qualitative error analysis, so that conclusions speak to clinical deployment, not only in-dataset benchmark scores.

1.5.2 Objectives

- Define a clinically relevant patch-level task on H&E lymph-node tissue (metastasis present vs absent in the central region) and adopt PCam and CAMELYON17 with their official splits, documenting label meaning, patch size, and in-domain vs out-of-domain evaluation.
- Establish a unified, traceable preprocessing pipeline, with fixed project standards and documented adaptive steps, applicable across datasets.
- Implement data-integrity controls including exact deduplication on PCam, quality screening on both corpora, cross-dataset overlap checks, and auditable manifests so that performance estimates are not inflated by duplicates or leakage.
- Characterize both datasets through exploratory and audit workflows (class balance, tissue and colour statistics, quality-filter impact, split-wise summaries) to support methodological transparency in the report.
- Compare stain-handling strategies under matched conditions, select a primary normalizer for full-corpus preprocessing, and document the rationale (benchmark CNN and/or prespecified metrics).
- Train and evaluate a conventional CNN baseline on PCam to quantify performance before foundation-model transfer.

- Develop Virchow2-based classifiers with a frozen encoder and trainable head, consistent input preparation, and reproducible experiment artefacts (weights, logs, predictions).
- Conduct bidirectional cross-domain evaluation by training on PCam → testing on CAMELYON17 and CAMELYON17 → PCam, and reporting in-domain and external-domain performance plus transfer degradation in both directions.
- Assess predictive reliability using calibration-focused metrics and post-hoc temperature scaling fitted on validation only, separating ranking performance (ROC/PR-AUC) from probability quality.
- Incorporate uncertainty-aware evaluation to support analysis of high-uncertainty correct and high-uncertainty error cases alongside standard mistakes. THE MC THING WAS IT DONE ?
- Perform structured qualitative error analysis using predefined buckets, a human review checklist, and exported review packages for case-based interpretation.
- Document reproducibility and experimental governance through versioned scripts, run manifests, naming conventions for conditions, and explicit separation of curation vs per-patch preprocessing so that the pipeline can be replicated or extended.

In sum, this study will test whether quality-aware preprocessing with a pathology foundation model improves metastasis detection across institutions, with more stable performance under staining and scanner differences and more reliable, uncertainty-informed probabilities for clinical use.

2. Literature Review

2.1 The Clinical Necessity and Evolution of Automated Pathology

The clinical case for automated pathology starts with workload and access. Pathology is central to cancer diagnosis and treatment, but global workforce pressure, rising cancer incidence, and unequal access to pathology services have made timely reporting harder to sustain [39], [40]. Studies have shown breast cancer subtyping accuracy rates as low as 57.08%, while prostate cancer grading agreement has been reported at only 57.90% [9]. Such discrepancies directly impact the TNM classification system and, consequently, patient treatment decisions.

The first automated approaches in pathology did not begin with deep learning; they began with morphometry and computer-assisted image analysis. Early work in the 1990s argued that image analysis could provide an objective basis for histopathologic decision-making, and by the late 1990s and early 2000s, reviews of automated histometry and quantitative histology described systems that segmented tissue, measured nuclear and glandular features, and used those measurements to support diagnosis [41], [42]. During this Pre- deep learning era, computational pathology relied on traditional machine learning, where researchers manually extracted morphological and texture features, such as Local Binary Patterns (LBP), Scale-Invariant Feature Transform (SIFT), and Grey-Level Co-occurrence Matrices (GLCM), and classified them using Support Vector Machines (SVMs), Random Forests, and other algorithms[43].

While these methods demonstrated that automated detection was possible across various cancer types (breast, colorectal, prostate, gastric), they were heavily dependent on human intuition for feature selection and struggled to capture the immense heterogeneity of tumor microenvironments[44]. By the time digital whole-slide images became practical, the field had already moved toward computer-aided diagnosis as a formal research area. Reviews from the 2000s and 2010s describe how digital scanners, image analysis, and machine learning turned histopathology into a true computational problem, setting up the later transition to slide-level deep learning and benchmark datasets [45], [46].

2.2 The Gigapixel Challenge and Convolutional Neural Networks

The transition from traditional microscopy to digital pathology introduced a fundamental computational challenge. A single whole-slide image (WSI) digitized at high resolution (e.g., 240 nm/pixel) is a multi-gigapixel image that cannot be loaded into standard GPU memory in its entirety [47]. For context, the CAMELYON dataset alone, comprising 1,399 WSIs from breast cancer patients, consumes roughly 2.95 terabytes of storage [21]. Because processing these images whole is computationally infeasible, researchers developed patch-based paradigms: WSIs are tiled into small patches (typically 256×256 pixels), classified individually, and aggregated into slide-level predictions

The Camelyon Grand Challenges were established to catalyze automated solutions. CAMELYON16 demonstrated that the best deep learning algorithms could achieve an AUC of 0.994, outperforming pathologists under simulated time constraints [8]. CAMELYON17 extended this to patient-level pN-staging across five centers, highlighting challenges in detecting Isolated Tumor Cells (ITCs). To facilitate patch-level research, PatchCamelyon (PCam) was introduced, a dataset of 327,680 labeled tiles extracted from CAMELYON16 slides. PCam has since become a standard benchmark for histopathology feature extraction [37]. Recently, these datasets were systematically reprocessed with expert pixel annotations to re-evaluate modern foundation models and Multiple Instance Learning (MIL) methods [21].

The creation of these standardized benchmarks catalyzed a diverse range of algorithmic strategies. With the emergence of CNNs, models could automatically learn hierarchical features directly from raw pixel data, eliminating the need for handcrafted feature engineering. On PatchCamelyon (PCam), Aparicio et al. (2021) utilized a modified VGG16 to achieve an AUC of 0.90 [48], while hybrid models ensembling DenseNet169 and ResNet50 demonstrated that dense connectivity could effectively alleviate the vanishing gradient problem in deep pathology networks. Other specialized approaches, such as semi-supervised learning with pseudo-labels and supervised contrastive learning (e.g., HistopathAI), further improved performance and highlighted the generalizability challenges across multiple cancer types [49]. Despite these advances, researchers identified significant domain-specific limitations in standard CNNs. First, histopathology images exhibit inherent rotational and reflectional symmetries; tissue appears equally valid at any orientation, yet standard CNNs often waste parameter capacity learning these variations redundantly. To address this, P4M-DenseNet, a rotation-equivariant CNN, was introduced alongside the PCam benchmark to explicitly encode these symmetries, resulting in more stable predictions [50]. This concept was further extended with Dense Steerable Filter CNNs, achieving state-of-the-art results with significantly fewer parameters [51].

Furthermore, isolated patch analysis often lacks the global spatial context necessary for accurate diagnosis. The MMSEN (Multi-scale Multi-head Self-attention Ensemble Network) addressed this by integrating self-attention mechanisms to guide the model toward critical tumor-related features, reaching a ROC-AUC of 99.01% on PCam [52], [53].

A pivotal shift occurred with the adoption of ImageNet-pretrained networks. Despite the large visual domain gap between natural photographs and microscopy, fine-tuning architectures like Inception-V3 and ResNet50 proved highly effective, with ResNet50 achieving 97.50% accuracy on breast histology subtypes [54]. Comparative studies confirmed that fine-tuned networks substantially outperform those trained from scratch [55]. However, the visual dissimilarity between natural images and histopathology remained a persistent concern. Comparisons between domain-specific and ImageNet weights revealed that pathology-specific pretraining yields higher performance and better feature reuse, foreshadowing the industry-wide shift toward large-scale pathology foundation models [56].

2.3 Preprocessing: Stain Normalization and Color Augmentation

A fundamental challenge in computational pathology, not only in breast cancer metastasis detection but across organ types and cancer subtypes, is that H&E stained slides exhibit substantial color variation across institutions, arising from differences in staining protocols, reagent brands, tissue preparation, section thickness, and scanner hardware. This variation degrades model performance on unseen institutions, a problem known as domain shift.

The Macenko method works by decomposing H&E images into stain vectors via singular value decomposition in optical density space, then mapping to a reference template [25]. This method remains widely used for its simplicity. The Vahadane method extended this using sparse non-negative matrix factorization, which better preserves tissue morphology. Comparative evaluations show Vahadane achieves higher structural fidelity (SSIM), while Macenko provides more consistent chromatic distributions [26]. At the whole-slide level, WSICS is a fully automated algorithm that aligns chromatic and density distributions per stain component to a template WSI, achieving the lowest color variation across 125 WSIs from 5 laboratories [57]. A comprehensive review in Information Fusion, grouping methods into four categories, showed that structure-preserving unified transformation-based methods consistently outperform alternatives [58].

Stain augmentation, normalization, and a neural network-based normalization were compared on heterogeneous data from 9 laboratories and 4 organs. Their key finding was that extensive stain augmentation in H&E color space was more effective than normalization alone for CNN generalization [27]. Augmentation directly in H&E color space (decomposing via color

deconvolution, modifying H and E channels stochastically, then projecting back to RGB) significantly improved robustness to unseen stain variations [59].

Six configurations combining Macenko and Vahadane normalization with color augmentation on PCam were compared. Macenko combined with color augmentation improved accuracy by 1.59% and F1 by 2.50% over baseline, modest gains reflecting PCam's relatively homogeneous origin, but consistent [60]. CAGAN is a color adaptive generative network combining supervised and unsupervised learning, reporting 5–10% improvement on CAMELYON16, CAMELYON17, and other datasets over baselines without normalization [61]. Bayesian blind color deconvolution improved AUC from 0.9491 to 0.9656 on Camelyon-16 and from 0.9279 to 0.9541 on Camelyon-17 [62]. Color normalization improved retrieval accuracy to 97% on CAMELYON17, a 14% increase [63]. Aggregating predictions from the two best normalization methods with augmentation improved external-set AUC from 0.860 to 0.892, confirming preprocessing benefits are most pronounced when generalizing across institutions [64].

HistAuGAN is a structure-preserving multi-domain stain augmentation using disentangled representations. When evaluated on Camelyon17, it outperformed conventional color augmentation and mitigated batch effects [65]. Learnable preprocessing operations trainable end-to-end with classification networks were also proposed [66]. Domain-specific modifications to RandAugment for H&E achieved AUC of 0.964 on Camelyon17 versus 0.958 for manual augmentation[67], [68].

AIDA (Adversarial Fourier-based Domain Adaptation) was proposed, which significantly outperformed both color normalization and standard adversarial approaches across four cancer types from multiple centers, suggesting stain normalization alone is insufficient for full cross-center generalization [69]. It was similarly found that adversarial training yielded significant improvements over color-normalized approaches. Additionally, combining stain normalization with segmentation-driven patch extraction was shown to improve accuracy by 17–18% for prostate and breast cancer detection [70].

Most normalization methods require selecting a reference image, introducing subjectivity. GAN-based methods risk mode collapse or information loss. The interaction between normalization and modern foundation models remains underexplored , whether models pretrained on millions of

diverse slides already encode stain-invariant features, rendering explicit normalization redundant, is an open question.

2.4 Self-Supervised Learning and Pathology Foundation Models

The scarcity of labeled histopathology data, where annotation requires expert pathologists, motivated self-supervised learning (SSL) approaches that could leverage the vast quantities of unlabeled WSIs available across institutions. This "annotation bottleneck" was a persistent limitation of fully supervised approaches. The power of SSL for clinical applications was demonstrated by utilizing the Bootstrap Your Own Latent (BYOL) framework to accurately identify eyelid malignant melanoma using only a fraction of the labeled data previously required, confirming that self-supervised features transfer effectively to rare pathology tasks [71].

when contrastive self-supervised learning across 60 histopathology datasets without labels is applied, it was found that histopathology-pretrained networks outperformed ImageNet-pretrained ones by up to 7.5% in accuracy and 8.9% in F1 [72].

CTransPath is a hybrid CNN-Transformer backbone pretrained using semantically-relevant contrastive learning (SRCL). It achieved state-of-the-art on nine public datasets spanning five task types, outperforming both ImageNet-supervised and self-supervised pretraining [73].

UNI is pretrained on over 100 million images from 100,000+ diagnostic WSIs across 20 tissue types. Evaluated on 34 tasks, UNI outperformed prior state-of-the-art and demonstrated resolution-agnostic classification and few-shot slide classification [74].

CONCH is a visual-language foundation model trained on over 1.17 million image-caption pairs. CONCH achieved state-of-the-art on 14 diverse benchmarks spanning classification, segmentation, captioning, and retrieval [75].

Prov-GigaPath is a whole-slide pathology foundation model pretrained on 1.3 billion 256×256 pathology image tiles from 171,189 WSIs covering 31 tissue types. Using a novel vision transformer architecture adapted from LongNet for ultra-large-context modeling, GigaPath achieved state-of-the-art on 25 out of 26 tasks in their benchmark [76].

Virchow is trained on 1.5 million WSIs from 100,000 patients. It achieved 0.95 specimen-level AUC for pan-cancer detection across nine common and seven rare cancers, and showed that with limited labeled data, Virchow-based detectors matched tissue-specific clinical-grade models [77].

Virchow2 (632M parameters), Virchow2G (1.9B), and a distilled 22M variant, trained on 3.1 million WSIs. They achieved state-of-the-art on 12 tile-level tasks and showed that combining domain-specific methods, data scale, and model scale yields the best performance [78].

The rapid proliferation of foundation models created urgent need for independent evaluation. When 19 models across 13 cohorts (6,818 patients, 9,528 slides) were benchmarked, CONCH yielded the highest performance, Virchow2 close second. An ensemble of CONCH and Virchow2 outperformed individual models in 55% of tasks [79]. Furthermore, 31 models across 41 tasks were benchmarked, finding Virchow2 highest overall and that model/data size did not consistently correlate with performance [79]. PathBench (presented across 15,888 WSIs from 10 hospitals) found Virchow2 and H-Optimus-1 the most effective [80].

2.6 Multiple Instance Learning for Whole-Slide Prediction

While patch-level classification is the building block, clinical decisions require slide-level or patient-level predictions. MIL treats each slide as a "bag" of patch instances. A proposed MIL-CNN framework achieved up to 89.52% accuracy for breast tumors [81]. A Prototypical MIL using unsupervised clustering achieved AUC of 88.2% on Camelyon16 and outperforming existing MIL methods for micro-metastases [82]. Modern pipelines now use foundation model features (from UNI, GigaPath, CTransPath, etc.) fed into advanced MIL aggregators such as ABMIL, CLAM, or TransMIL to generate slide-level predictions [83].

2.7 Gaps, Limitations, and Future Directions

Despite considerable progress, several gaps remain. The interaction between preprocessing pipelines and foundation model features is largely untested, whether explicit normalization adds benefit when using models pretrained on millions of diverse slides is an open question. Independent benchmarks may suffer from data leakage with public datasets. ITCs remain the most challenging subtype (detection below 40%). The computational cost of the largest models limits accessibility, though models trained on far fewer WSIs can match Virchow2 [84]. Most work focuses on H&E stains; generalization to other protocols is underexplored. Finally,

the optimal combination of adaptive preprocessing, domain-specific augmentation, and foundation model transfer learning for cross-center generalizability, the central aim of the present work, has not been systematically investigated.

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